

# Solução do Problema scsd8 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

## 1 Informações do Problema

### Informações do Problema:

- Nome: scsd8
- Número de Variáveis: 2750
- Número de Restrições: 397
- Inviabilidade Primal: 2.120e-15
- Inviabilidade Dual: 1.599e-13
- Valor Primal: 9.050e+02
- Valor Dual: 9.050e+02
- Gap: 6.243e-19
- Número de Iterações: 18

## 2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema scsd8

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 1            | 0.000e+00        | 1            | 2.000e+00                |
| 2            | 0.000e+00        | 2            | 0.000e+00                |
| 3            | 0.000e+00        | 3            | 4.000e+00                |
| 4            | 0.000e+00        | 4            | 0.000e+00                |
| 5            | 0.000e+00        | 5            | 6.000e+00                |
| 6            | 0.000e+00        | 6            | 0.000e+00                |
| 7            | 0.000e+00        | 7            | 8.000e+00                |
| 8            | 0.000e+00        | 8            | 0.000e+00                |
| 9            | 0.000e+00        | 9            | 2.000e+00                |
| 10           | 0.000e+00        | 10           | 0.000e+00                |
| 11           | 0.000e+00        | 11           | 2.357e+00                |
| 12           | 0.000e+00        | 12           | 4.714e-01                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 13           | 0.000e+00        | 13           | 2.981e+00                |
| 14           | 0.000e+00        | 14           | 1.491e+00                |
| 15           | 0.000e+00        | 15           | 4.427e+00                |
| 16           | 0.000e+00        | 16           | 1.897e+00                |
| 17           | 0.000e+00        | 17           | 6.306e+00                |
| 18           | 0.000e+00        | 18           | 1.940e+00                |
| 19           | 0.000e+00        | 19           | 2.000e+00                |
| 20           | 0.000e+00        | 20           | 0.000e+00                |
| 21           | 0.000e+00        | 21           | 4.000e+00                |
| 22           | 0.000e+00        | 22           | 0.000e+00                |
| 23           | 0.000e+00        | 23           | 6.000e+00                |
| 24           | 0.000e+00        | 24           | 0.000e+00                |
| 25           | 0.000e+00        | 25           | 2.828e+00                |
| 26           | 0.000e+00        | 26           | 0.000e+00                |
| 27           | 0.000e+00        | 27           | 2.000e+00                |
| 28           | 1.000e+00        | 28           | 0.000e+00                |
| 29           | 0.000e+00        | 29           | 1.886e+00                |
| 30           | 0.000e+00        | 30           | 9.428e-01                |
| 31           | 0.000e+00        | 31           | 3.280e+00                |
| 32           | 0.000e+00        | 32           | 1.193e+00                |
| 33           | 0.000e+00        | 33           | 5.060e+00                |
| 34           | 0.000e+00        | 34           | 1.265e+00                |
| 35           | 0.000e+00        | 35           | 2.000e+00                |
| 36           | 0.000e+00        | 36           | 0.000e+00                |
| 37           | 0.000e+00        | 37           | 4.000e+00                |
| 38           | 0.000e+00        | 38           | 0.000e+00                |
| 39           | 0.000e+00        | 39           | 4.174e+00                |
| 40           | 0.000e+00        | 40           | 2.981e-01                |
| 41           | 0.000e+00        | 41           | 2.828e+00                |
| 42           | 2.357e-01        | 42           | 0.000e+00                |
| 43           | 0.000e+00        | 43           | 2.000e+00                |
| 44           | 6.667e-01        | 44           | 0.000e+00                |
| 45           | 0.000e+00        | 45           | 2.828e+00                |
| 46           | 2.357e-01        | 46           | 0.000e+00                |
| 47           | 0.000e+00        | 47           | 4.174e+00                |
| 48           | 0.000e+00        | 48           | 2.981e-01                |
| 49           | 0.000e+00        | 49           | 2.000e+00                |
| 50           | 0.000e+00        | 50           | 0.000e+00                |
| 51           | 0.000e+00        | 51           | 5.060e+00                |
| 52           | 0.000e+00        | 52           | 1.265e+00                |
| 53           | 0.000e+00        | 53           | 3.280e+00                |
| 54           | 0.000e+00        | 54           | 1.193e+00                |
| 55           | 0.000e+00        | 55           | 1.886e+00                |
| 56           | 0.000e+00        | 56           | 9.428e-01                |
| 57           | 0.000e+00        | 57           | 2.000e+00                |
| 58           | 1.000e+00        | 58           | 0.000e+00                |
| 59           | 0.000e+00        | 59           | 2.828e+00                |
| 60           | 0.000e+00        | 60           | 0.000e+00                |
| 61           | 0.000e+00        | 61           | 6.306e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 62           | 0.000e+00        | 62           | 1.940e+00                |
| 63           | 0.000e+00        | 63           | 4.427e+00                |
| 64           | 0.000e+00        | 64           | 1.897e+00                |
| 65           | 0.000e+00        | 65           | 2.981e+00                |
| 66           | 0.000e+00        | 66           | 1.491e+00                |
| 67           | 0.000e+00        | 67           | 2.357e+00                |
| 68           | 0.000e+00        | 68           | 4.714e-01                |
| 69           | 0.000e+00        | 69           | 2.000e+00                |
| 70           | 0.000e+00        | 70           | 0.000e+00                |
| 71           | 0.000e+00        | 71           | 1.333e+00                |
| 72           | 0.000e+00        | 72           | 6.667e-01                |
| 73           | 0.000e+00        | 73           | 2.000e+00                |
| 74           | 0.000e+00        | 74           | 2.000e+00                |
| 75           | 0.000e+00        | 75           | 2.667e+00                |
| 76           | 0.000e+00        | 76           | 3.333e+00                |
| 77           | 0.000e+00        | 77           | 4.000e+00                |
| 78           | 0.000e+00        | 78           | 4.000e+00                |
| 79           | 0.000e+00        | 79           | 2.000e+00                |
| 80           | 0.000e+00        | 80           | 0.000e+00                |
| 81           | 0.000e+00        | 81           | 9.428e-01                |
| 82           | 0.000e+00        | 82           | 1.886e+00                |
| 83           | 0.000e+00        | 83           | 1.193e+00                |
| 84           | 0.000e+00        | 84           | 3.280e+00                |
| 85           | 0.000e+00        | 85           | 2.530e+00                |
| 86           | 0.000e+00        | 86           | 3.795e+00                |
| 87           | 0.000e+00        | 87           | 3.072e+00                |
| 88           | 0.000e+00        | 88           | 5.174e+00                |
| 89           | 0.000e+00        | 89           | 6.667e-01                |
| 90           | 0.000e+00        | 90           | 1.333e+00                |
| 91           | 0.000e+00        | 91           | 1.333e+00                |
| 92           | 0.000e+00        | 92           | 2.667e+00                |
| 93           | 0.000e+00        | 93           | 2.667e+00                |
| 94           | 0.000e+00        | 94           | 3.333e+00                |
| 95           | 0.000e+00        | 95           | 2.828e+00                |
| 96           | 2.357e-01        | 96           | 0.000e+00                |
| 97           | 0.000e+00        | 97           | 2.000e+00                |
| 98           | 1.000e+00        | 98           | 0.000e+00                |
| 99           | 0.000e+00        | 99           | 9.428e-01                |
| 100          | 0.000e+00        | 100          | 1.886e+00                |
| 101          | 0.000e+00        | 101          | 2.087e+00                |
| 102          | 0.000e+00        | 102          | 2.385e+00                |
| 103          | 0.000e+00        | 103          | 2.530e+00                |
| 104          | 0.000e+00        | 104          | 3.795e+00                |
| 105          | 0.000e+00        | 105          | 6.667e-01                |
| 106          | 0.000e+00        | 106          | 1.333e+00                |
| 107          | 0.000e+00        | 107          | 2.000e+00                |
| 108          | 0.000e+00        | 108          | 2.000e+00                |
| 109          | 0.000e+00        | 109          | 2.981e+00                |
| 110          | 0.000e+00        | 110          | 1.491e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 111          | 0.000e+00        | 111          | 2.828e+00                |
| 112          | 0.000e+00        | 112          | -7.327e-15               |
| 113          | 0.000e+00        | 113          | 2.000e+00                |
| 114          | 6.667e-01        | 114          | 0.000e+00                |
| 115          | 0.000e+00        | 115          | 2.828e+00                |
| 116          | 0.000e+00        | 116          | 0.000e+00                |
| 117          | 0.000e+00        | 117          | 2.981e+00                |
| 118          | 0.000e+00        | 118          | 1.491e+00                |
| 119          | 0.000e+00        | 119          | 1.333e+00                |
| 120          | 0.000e+00        | 120          | 6.667e-01                |
| 121          | 0.000e+00        | 121          | 2.530e+00                |
| 122          | 0.000e+00        | 122          | 3.795e+00                |
| 123          | 0.000e+00        | 123          | 2.087e+00                |
| 124          | 0.000e+00        | 124          | 2.385e+00                |
| 125          | 0.000e+00        | 125          | 9.428e-01                |
| 126          | 0.000e+00        | 126          | 1.886e+00                |
| 127          | 0.000e+00        | 127          | 2.000e+00                |
| 128          | 1.000e+00        | 128          | 0.000e+00                |
| 129          | 0.000e+00        | 129          | 2.828e+00                |
| 130          | 2.357e-01        | 130          | 0.000e+00                |
| 131          | 0.000e+00        | 131          | 3.072e+00                |
| 132          | 0.000e+00        | 132          | 5.174e+00                |
| 133          | 0.000e+00        | 133          | 2.530e+00                |
| 134          | 0.000e+00        | 134          | 3.795e+00                |
| 135          | 0.000e+00        | 135          | 1.193e+00                |
| 136          | 0.000e+00        | 136          | 3.280e+00                |
| 137          | 0.000e+00        | 137          | 9.428e-01                |
| 138          | 0.000e+00        | 138          | 1.886e+00                |
| 139          | 0.000e+00        | 139          | 2.000e+00                |
| 140          | 0.000e+00        | 140          | 0.000e+00                |
| 141          | 1.667e-01        | 141          | 0.000e+00                |
| 142          | 0.000e+00        | 142          | 2.000e+00                |
| 143          | 0.000e+00        | 143          | 6.667e-01                |
| 144          | 0.000e+00        | 144          | 3.333e+00                |
| 145          | 0.000e+00        | 145          | 1.333e+00                |
| 146          | 0.000e+00        | 146          | 4.667e+00                |
| 147          | 0.000e+00        | 147          | 1.333e+00                |
| 148          | 0.000e+00        | 148          | 6.667e+00                |
| 149          | 0.000e+00        | 149          | 2.000e+00                |
| 150          | 1.667e-01        | 150          | 0.000e+00                |
| 151          | 0.000e+00        | 151          | 3.331e-15                |
| 152          | 0.000e+00        | 152          | 2.828e+00                |
| 153          | 0.000e+00        | 153          | 0.000e+00                |
| 154          | 0.000e+00        | 154          | 4.472e+00                |
| 155          | 0.000e+00        | 155          | 1.265e+00                |
| 156          | 0.000e+00        | 156          | 5.060e+00                |
| 157          | 0.000e+00        | 157          | 1.779e+00                |
| 158          | 0.000e+00        | 158          | 6.468e+00                |
| 159          | 0.000e+00        | 159          | 6.667e-01                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 160          | 0.000e+00        | 160          | 1.333e+00                |
| 161          | 0.000e+00        | 161          | 1.333e+00                |
| 162          | 0.000e+00        | 162          | 2.667e+00                |
| 163          | 0.000e+00        | 163          | 1.333e+00                |
| 164          | 0.000e+00        | 164          | 4.667e+00                |
| 165          | 0.000e+00        | 165          | 2.828e+00                |
| 166          | 2.357e-01        | 166          | 0.000e+00                |
| 167          | 0.000e+00        | 167          | 2.000e+00                |
| 168          | 8.333e-01        | 168          | 0.000e+00                |
| 169          | 0.000e+00        | 169          | 9.428e-01                |
| 170          | 0.000e+00        | 170          | 1.886e+00                |
| 171          | 0.000e+00        | 171          | 2.087e+00                |
| 172          | 0.000e+00        | 172          | 2.385e+00                |
| 173          | 0.000e+00        | 173          | 2.530e+00                |
| 174          | 0.000e+00        | 174          | 3.795e+00                |
| 175          | 0.000e+00        | 175          | 6.667e-01                |
| 176          | 0.000e+00        | 176          | 1.333e+00                |
| 177          | 0.000e+00        | 177          | 6.667e-01                |
| 178          | 0.000e+00        | 178          | 3.333e+00                |
| 179          | 0.000e+00        | 179          | 2.981e+00                |
| 180          | 0.000e+00        | 180          | 1.491e+00                |
| 181          | 0.000e+00        | 181          | 2.828e+00                |
| 182          | 1.963e-16        | 182          | 0.000e+00                |
| 183          | 0.000e+00        | 183          | 2.000e+00                |
| 184          | 6.667e-01        | 184          | 0.000e+00                |
| 185          | 0.000e+00        | 185          | 2.828e+00                |
| 186          | 1.963e-16        | 186          | 0.000e+00                |
| 187          | 0.000e+00        | 187          | 2.981e+00                |
| 188          | 0.000e+00        | 188          | 1.491e+00                |
| 189          | 1.667e-01        | 189          | 0.000e+00                |
| 190          | 0.000e+00        | 190          | 2.000e+00                |
| 191          | 0.000e+00        | 191          | 2.530e+00                |
| 192          | 0.000e+00        | 192          | 3.795e+00                |
| 193          | 0.000e+00        | 193          | 2.087e+00                |
| 194          | 0.000e+00        | 194          | 2.385e+00                |
| 195          | 0.000e+00        | 195          | 9.428e-01                |
| 196          | 0.000e+00        | 196          | 1.886e+00                |
| 197          | 0.000e+00        | 197          | 2.000e+00                |
| 198          | 8.333e-01        | 198          | 0.000e+00                |
| 199          | 0.000e+00        | 199          | 2.828e+00                |
| 200          | 2.357e-01        | 200          | 0.000e+00                |
| 201          | 0.000e+00        | 201          | 1.779e+00                |
| 202          | 0.000e+00        | 202          | 6.468e+00                |
| 203          | 0.000e+00        | 203          | 1.265e+00                |
| 204          | 0.000e+00        | 204          | 5.060e+00                |
| 205          | 0.000e+00        | 205          | 6.084e-14                |
| 206          | 0.000e+00        | 206          | 4.472e+00                |
| 207          | 0.000e+00        | 207          | -2.220e-16               |
| 208          | 0.000e+00        | 208          | 2.828e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 209          | 0.000e+00        | 209          | 2.000e+00                |
| 210          | 1.667e-01        | 210          | 0.000e+00                |
| 211          | 0.000e+00        | 211          | 0.000e+00                |
| 212          | 0.000e+00        | 212          | 2.000e+00                |
| 213          | 0.000e+00        | 213          | 6.667e-01                |
| 214          | 0.000e+00        | 214          | 3.333e+00                |
| 215          | 0.000e+00        | 215          | 1.333e+00                |
| 216          | 0.000e+00        | 216          | 4.667e+00                |
| 217          | 0.000e+00        | 217          | 1.333e+00                |
| 218          | 0.000e+00        | 218          | 6.667e+00                |
| 219          | 0.000e+00        | 219          | 2.000e+00                |
| 220          | 5.000e-01        | 220          | 0.000e+00                |
| 221          | 2.357e-01        | 221          | 0.000e+00                |
| 222          | 0.000e+00        | 222          | 2.828e+00                |
| 223          | 0.000e+00        | 223          | 4.130e-14                |
| 224          | 0.000e+00        | 224          | 4.472e+00                |
| 225          | 0.000e+00        | 225          | 1.265e+00                |
| 226          | 0.000e+00        | 226          | 5.060e+00                |
| 227          | 0.000e+00        | 227          | 1.779e+00                |
| 228          | 0.000e+00        | 228          | 6.468e+00                |
| 229          | 0.000e+00        | 229          | 6.667e-01                |
| 230          | 0.000e+00        | 230          | 1.333e+00                |
| 231          | 0.000e+00        | 231          | 1.333e+00                |
| 232          | 0.000e+00        | 232          | 2.667e+00                |
| 233          | 0.000e+00        | 233          | 1.333e+00                |
| 234          | 0.000e+00        | 234          | 4.667e+00                |
| 235          | 0.000e+00        | 235          | 2.828e+00                |
| 236          | 1.963e-16        | 236          | 0.000e+00                |
| 237          | 0.000e+00        | 237          | 2.000e+00                |
| 238          | 8.333e-01        | 238          | 0.000e+00                |
| 239          | 0.000e+00        | 239          | 9.428e-01                |
| 240          | 0.000e+00        | 240          | 1.886e+00                |
| 241          | 0.000e+00        | 241          | 2.087e+00                |
| 242          | 0.000e+00        | 242          | 2.385e+00                |
| 243          | 0.000e+00        | 243          | 2.530e+00                |
| 244          | 0.000e+00        | 244          | 3.795e+00                |
| 245          | 0.000e+00        | 245          | 6.667e-01                |
| 246          | 0.000e+00        | 246          | 1.333e+00                |
| 247          | 0.000e+00        | 247          | 6.667e-01                |
| 248          | 0.000e+00        | 248          | 3.333e+00                |
| 249          | 0.000e+00        | 249          | 2.981e+00                |
| 250          | 0.000e+00        | 250          | 1.491e+00                |
| 251          | 0.000e+00        | 251          | 2.828e+00                |
| 252          | 0.000e+00        | 252          | 3.886e-14                |
| 253          | 0.000e+00        | 253          | 2.000e+00                |
| 254          | 6.667e-01        | 254          | 0.000e+00                |
| 255          | 0.000e+00        | 255          | 2.828e+00                |
| 256          | 0.000e+00        | 256          | -1.443e-14               |
| 257          | 0.000e+00        | 257          | 2.981e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 258          | 0.000e+00        | 258          | 1.491e+00                |
| 259          | 1.667e-01        | 259          | 0.000e+00                |
| 260          | 0.000e+00        | 260          | 2.000e+00                |
| 261          | 0.000e+00        | 261          | 2.530e+00                |
| 262          | 0.000e+00        | 262          | 3.795e+00                |
| 263          | 0.000e+00        | 263          | 2.087e+00                |
| 264          | 0.000e+00        | 264          | 2.385e+00                |
| 265          | 0.000e+00        | 265          | 9.428e-01                |
| 266          | 0.000e+00        | 266          | 1.886e+00                |
| 267          | 0.000e+00        | 267          | 2.000e+00                |
| 268          | 6.667e-01        | 268          | 0.000e+00                |
| 269          | 0.000e+00        | 269          | 2.828e+00                |
| 270          | 2.357e-01        | 270          | 0.000e+00                |
| 271          | 0.000e+00        | 271          | 1.779e+00                |
| 272          | 0.000e+00        | 272          | 6.468e+00                |
| 273          | 0.000e+00        | 273          | 1.265e+00                |
| 274          | 0.000e+00        | 274          | 5.060e+00                |
| 275          | 0.000e+00        | 275          | -2.975e-14               |
| 276          | 0.000e+00        | 276          | 4.472e+00                |
| 277          | 0.000e+00        | 277          | -2.220e-16               |
| 278          | 0.000e+00        | 278          | 2.828e+00                |
| 279          | 0.000e+00        | 279          | 2.000e+00                |
| 280          | 3.333e-01        | 280          | 0.000e+00                |
| 281          | 0.000e+00        | 281          | 2.998e-15                |
| 282          | 0.000e+00        | 282          | 2.000e+00                |
| 283          | 0.000e+00        | 283          | 6.667e-01                |
| 284          | 0.000e+00        | 284          | 3.333e+00                |
| 285          | 0.000e+00        | 285          | 1.333e+00                |
| 286          | 0.000e+00        | 286          | 4.667e+00                |
| 287          | 0.000e+00        | 287          | 1.333e+00                |
| 288          | 0.000e+00        | 288          | 6.667e+00                |
| 289          | 0.000e+00        | 289          | 2.000e+00                |
| 290          | 5.000e-01        | 290          | 0.000e+00                |
| 291          | 0.000e+00        | 291          | 1.754e-14                |
| 292          | 0.000e+00        | 292          | 2.828e+00                |
| 293          | 1.552e-16        | 293          | 0.000e+00                |
| 294          | 0.000e+00        | 294          | 4.472e+00                |
| 295          | 0.000e+00        | 295          | 1.265e+00                |
| 296          | 0.000e+00        | 296          | 5.060e+00                |
| 297          | 0.000e+00        | 297          | 1.779e+00                |
| 298          | 0.000e+00        | 298          | 6.468e+00                |
| 299          | 0.000e+00        | 299          | 6.667e-01                |
| 300          | 0.000e+00        | 300          | 1.333e+00                |
| 301          | 0.000e+00        | 301          | 1.333e+00                |
| 302          | 0.000e+00        | 302          | 2.667e+00                |
| 303          | 0.000e+00        | 303          | 1.333e+00                |
| 304          | 0.000e+00        | 304          | 4.667e+00                |
| 305          | 0.000e+00        | 305          | 2.828e+00                |
| 306          | 2.357e-01        | 306          | 0.000e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 307          | 0.000e+00        | 307          | 2.000e+00                |
| 308          | 5.000e-01        | 308          | 0.000e+00                |
| 309          | 0.000e+00        | 309          | 9.428e-01                |
| 310          | 0.000e+00        | 310          | 1.886e+00                |
| 311          | 0.000e+00        | 311          | 2.087e+00                |
| 312          | 0.000e+00        | 312          | 2.385e+00                |
| 313          | 0.000e+00        | 313          | 2.530e+00                |
| 314          | 0.000e+00        | 314          | 3.795e+00                |
| 315          | 0.000e+00        | 315          | 6.667e-01                |
| 316          | 0.000e+00        | 316          | 1.333e+00                |
| 317          | 0.000e+00        | 317          | 6.667e-01                |
| 318          | 0.000e+00        | 318          | 3.333e+00                |
| 319          | 0.000e+00        | 319          | 2.981e+00                |
| 320          | 0.000e+00        | 320          | 1.491e+00                |
| 321          | 0.000e+00        | 321          | 2.828e+00                |
| 322          | 0.000e+00        | 322          | 0.000e+00                |
| 323          | 0.000e+00        | 323          | 2.000e+00                |
| 324          | 6.667e-01        | 324          | 0.000e+00                |
| 325          | 0.000e+00        | 325          | 2.828e+00                |
| 326          | 0.000e+00        | 326          | 5.662e-14                |
| 327          | 0.000e+00        | 327          | 2.981e+00                |
| 328          | 0.000e+00        | 328          | 1.491e+00                |
| 329          | 1.667e-01        | 329          | 0.000e+00                |
| 330          | 0.000e+00        | 330          | 2.000e+00                |
| 331          | 0.000e+00        | 331          | 2.530e+00                |
| 332          | 0.000e+00        | 332          | 3.795e+00                |
| 333          | 0.000e+00        | 333          | 2.087e+00                |
| 334          | 0.000e+00        | 334          | 2.385e+00                |
| 335          | 0.000e+00        | 335          | 9.428e-01                |
| 336          | 0.000e+00        | 336          | 1.886e+00                |
| 337          | 0.000e+00        | 337          | 2.000e+00                |
| 338          | 5.000e-01        | 338          | 0.000e+00                |
| 339          | 0.000e+00        | 339          | 2.828e+00                |
| 340          | 2.357e-01        | 340          | 0.000e+00                |
| 341          | 0.000e+00        | 341          | 1.779e+00                |
| 342          | 0.000e+00        | 342          | 6.468e+00                |
| 343          | 0.000e+00        | 343          | 1.265e+00                |
| 344          | 0.000e+00        | 344          | 5.060e+00                |
| 345          | 0.000e+00        | 345          | -5.107e-14               |
| 346          | 0.000e+00        | 346          | 4.472e+00                |
| 347          | 0.000e+00        | 347          | -2.220e-16               |
| 348          | 0.000e+00        | 348          | 2.828e+00                |
| 349          | 0.000e+00        | 349          | 2.000e+00                |
| 350          | 5.000e-01        | 350          | 0.000e+00                |
| 351          | 1.667e-01        | 351          | 0.000e+00                |
| 352          | 0.000e+00        | 352          | 2.000e+00                |
| 353          | 0.000e+00        | 353          | 6.667e-01                |
| 354          | 0.000e+00        | 354          | 3.333e+00                |
| 355          | 0.000e+00        | 355          | 1.333e+00                |

Continua na próxima página



| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 356          | 0.000e+00        | 356          | 4.667e+00                |
| 357          | 0.000e+00        | 357          | 1.333e+00                |
| 358          | 0.000e+00        | 358          | 6.667e+00                |
| 359          | 0.000e+00        | 359          | 2.000e+00                |
| 360          | 6.667e-01        | 360          | 0.000e+00                |
| 361          | 0.000e+00        | 361          | 6.883e-15                |
| 362          | 0.000e+00        | 362          | 2.828e+00                |
| 363          | 0.000e+00        | 363          | -8.660e-14               |
| 364          | 0.000e+00        | 364          | 4.472e+00                |
| 365          | 0.000e+00        | 365          | 1.265e+00                |
| 366          | 0.000e+00        | 366          | 5.060e+00                |
| 367          | 0.000e+00        | 367          | 1.779e+00                |
| 368          | 0.000e+00        | 368          | 6.468e+00                |
| 369          | 0.000e+00        | 369          | 6.667e-01                |
| 370          | 0.000e+00        | 370          | 1.333e+00                |
| 371          | 0.000e+00        | 371          | 1.333e+00                |
| 372          | 0.000e+00        | 372          | 2.667e+00                |
| 373          | 0.000e+00        | 373          | 1.333e+00                |
| 374          | 0.000e+00        | 374          | 4.667e+00                |
| 375          | 0.000e+00        | 375          | 2.828e+00                |
| 376          | 2.357e-01        | 376          | 0.000e+00                |
| 377          | 0.000e+00        | 377          | 2.000e+00                |
| 378          | 3.333e-01        | 378          | 0.000e+00                |
| 379          | 0.000e+00        | 379          | 9.428e-01                |
| 380          | 0.000e+00        | 380          | 1.886e+00                |
| 381          | 0.000e+00        | 381          | 2.087e+00                |
| 382          | 0.000e+00        | 382          | 2.385e+00                |
| 383          | 0.000e+00        | 383          | 2.530e+00                |
| 384          | 0.000e+00        | 384          | 3.795e+00                |
| 385          | 0.000e+00        | 385          | 6.667e-01                |
| 386          | 0.000e+00        | 386          | 1.333e+00                |
| 387          | 0.000e+00        | 387          | 6.667e-01                |
| 388          | 0.000e+00        | 388          | 3.333e+00                |
| 389          | 0.000e+00        | 389          | 2.981e+00                |
| 390          | 0.000e+00        | 390          | 1.491e+00                |
| 391          | 0.000e+00        | 391          | 2.828e+00                |
| 392          | 1.963e-16        | 392          | 0.000e+00                |
| 393          | 0.000e+00        | 393          | 2.000e+00                |
| 394          | 6.667e-01        | 394          | 0.000e+00                |
| 395          | 0.000e+00        | 395          | 2.828e+00                |
| 396          | 0.000e+00        | 396          | 7.438e-14                |
| 397          | 0.000e+00        | 397          | 2.981e+00                |
| 398          | 0.000e+00        | 398          | 1.491e+00                |
| 399          | 1.667e-01        | 399          | 0.000e+00                |
| 400          | 0.000e+00        | 400          | 2.000e+00                |
| 401          | 0.000e+00        | 401          | 2.530e+00                |
| 402          | 0.000e+00        | 402          | 3.795e+00                |
| 403          | 0.000e+00        | 403          | 2.087e+00                |
| 404          | 0.000e+00        | 404          | 2.385e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 405          | 0.000e+00        | 405          | 9.428e-01                |
| 406          | 0.000e+00        | 406          | 1.886e+00                |
| 407          | 0.000e+00        | 407          | 2.000e+00                |
| 408          | 3.333e-01        | 408          | 0.000e+00                |
| 409          | 0.000e+00        | 409          | 2.828e+00                |
| 410          | 2.357e-01        | 410          | 0.000e+00                |
| 411          | 0.000e+00        | 411          | 1.779e+00                |
| 412          | 0.000e+00        | 412          | 6.468e+00                |
| 413          | 0.000e+00        | 413          | 1.265e+00                |
| 414          | 0.000e+00        | 414          | 5.060e+00                |
| 415          | 0.000e+00        | 415          | 3.952e-14                |
| 416          | 0.000e+00        | 416          | 4.472e+00                |
| 417          | 0.000e+00        | 417          | -3.775e-15               |
| 418          | 0.000e+00        | 418          | 2.828e+00                |
| 419          | 0.000e+00        | 419          | 2.000e+00                |
| 420          | 6.667e-01        | 420          | 0.000e+00                |
| 421          | 1.667e-01        | 421          | 0.000e+00                |
| 422          | 0.000e+00        | 422          | 2.000e+00                |
| 423          | 0.000e+00        | 423          | 6.667e-01                |
| 424          | 0.000e+00        | 424          | 3.333e+00                |
| 425          | 0.000e+00        | 425          | 1.333e+00                |
| 426          | 0.000e+00        | 426          | 4.667e+00                |
| 427          | 0.000e+00        | 427          | 1.333e+00                |
| 428          | 0.000e+00        | 428          | 6.667e+00                |
| 429          | 0.000e+00        | 429          | 2.000e+00                |
| 430          | 8.333e-01        | 430          | 0.000e+00                |
| 431          | 0.000e+00        | 431          | -2.220e-16               |
| 432          | 0.000e+00        | 432          | 2.828e+00                |
| 433          | 0.000e+00        | 433          | -1.554e-14               |
| 434          | 0.000e+00        | 434          | 4.472e+00                |
| 435          | 0.000e+00        | 435          | 1.265e+00                |
| 436          | 0.000e+00        | 436          | 5.060e+00                |
| 437          | 0.000e+00        | 437          | 1.779e+00                |
| 438          | 0.000e+00        | 438          | 6.468e+00                |
| 439          | 0.000e+00        | 439          | 6.667e-01                |
| 440          | 0.000e+00        | 440          | 1.333e+00                |
| 441          | 0.000e+00        | 441          | 1.333e+00                |
| 442          | 0.000e+00        | 442          | 2.667e+00                |
| 443          | 0.000e+00        | 443          | 1.333e+00                |
| 444          | 0.000e+00        | 444          | 4.667e+00                |
| 445          | 0.000e+00        | 445          | 2.828e+00                |
| 446          | 2.357e-01        | 446          | 0.000e+00                |
| 447          | 0.000e+00        | 447          | 2.000e+00                |
| 448          | 1.667e-01        | 448          | 0.000e+00                |
| 449          | 0.000e+00        | 449          | 9.428e-01                |
| 450          | 0.000e+00        | 450          | 1.886e+00                |
| 451          | 0.000e+00        | 451          | 2.087e+00                |
| 452          | 0.000e+00        | 452          | 2.385e+00                |
| 453          | 0.000e+00        | 453          | 2.530e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 454          | 0.000e+00        | 454          | 3.795e+00                |
| 455          | 0.000e+00        | 455          | 6.667e-01                |
| 456          | 0.000e+00        | 456          | 1.333e+00                |
| 457          | 0.000e+00        | 457          | 6.667e-01                |
| 458          | 0.000e+00        | 458          | 3.333e+00                |
| 459          | 0.000e+00        | 459          | 2.981e+00                |
| 460          | 0.000e+00        | 460          | 1.491e+00                |
| 461          | 0.000e+00        | 461          | 2.828e+00                |
| 462          | 0.000e+00        | 462          | 8.860e-14                |
| 463          | 0.000e+00        | 463          | 2.000e+00                |
| 464          | 6.667e-01        | 464          | 0.000e+00                |
| 465          | 0.000e+00        | 465          | 2.828e+00                |
| 466          | 0.000e+00        | 466          | 0.000e+00                |
| 467          | 0.000e+00        | 467          | 2.981e+00                |
| 468          | 0.000e+00        | 468          | 1.491e+00                |
| 469          | 1.667e-01        | 469          | 0.000e+00                |
| 470          | 0.000e+00        | 470          | 2.000e+00                |
| 471          | 0.000e+00        | 471          | 2.530e+00                |
| 472          | 0.000e+00        | 472          | 3.795e+00                |
| 473          | 0.000e+00        | 473          | 2.087e+00                |
| 474          | 0.000e+00        | 474          | 2.385e+00                |
| 475          | 0.000e+00        | 475          | 9.428e-01                |
| 476          | 0.000e+00        | 476          | 1.886e+00                |
| 477          | 0.000e+00        | 477          | 2.000e+00                |
| 478          | 1.667e-01        | 478          | 0.000e+00                |
| 479          | 0.000e+00        | 479          | 2.828e+00                |
| 480          | 2.357e-01        | 480          | 0.000e+00                |
| 481          | 0.000e+00        | 481          | 1.779e+00                |
| 482          | 0.000e+00        | 482          | 6.468e+00                |
| 483          | 0.000e+00        | 483          | 1.265e+00                |
| 484          | 0.000e+00        | 484          | 5.060e+00                |
| 485          | 0.000e+00        | 485          | -5.285e-14               |
| 486          | 0.000e+00        | 486          | 4.472e+00                |
| 487          | 0.000e+00        | 487          | -2.220e-16               |
| 488          | 0.000e+00        | 488          | 2.828e+00                |
| 489          | 0.000e+00        | 489          | 2.000e+00                |
| 490          | 8.333e-01        | 490          | 0.000e+00                |
| 491          | 1.667e-01        | 491          | 0.000e+00                |
| 492          | 0.000e+00        | 492          | 2.000e+00                |
| 493          | 0.000e+00        | 493          | 6.667e-01                |
| 494          | 0.000e+00        | 494          | 3.333e+00                |
| 495          | 0.000e+00        | 495          | 1.333e+00                |
| 496          | 0.000e+00        | 496          | 4.667e+00                |
| 497          | 0.000e+00        | 497          | 1.333e+00                |
| 498          | 0.000e+00        | 498          | 6.667e+00                |
| 499          | 0.000e+00        | 499          | 2.000e+00                |
| 500          | 1.000e+00        | 500          | 0.000e+00                |
| 501          | 0.000e+00        | 501          | 3.175e-14                |
| 502          | 0.000e+00        | 502          | 2.828e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 503          | 0.000e+00        | 503          | -1.026e-13               |
| 504          | 0.000e+00        | 504          | 4.472e+00                |
| 505          | 0.000e+00        | 505          | 1.265e+00                |
| 506          | 0.000e+00        | 506          | 5.060e+00                |
| 507          | 0.000e+00        | 507          | 1.779e+00                |
| 508          | 0.000e+00        | 508          | 6.468e+00                |
| 509          | 0.000e+00        | 509          | 6.667e-01                |
| 510          | 0.000e+00        | 510          | 1.333e+00                |
| 511          | 0.000e+00        | 511          | 1.333e+00                |
| 512          | 0.000e+00        | 512          | 2.667e+00                |
| 513          | 0.000e+00        | 513          | 1.333e+00                |
| 514          | 0.000e+00        | 514          | 4.667e+00                |
| 515          | 0.000e+00        | 515          | 2.828e+00                |
| 516          | 2.357e-01        | 516          | 0.000e+00                |
| 517          | 0.000e+00        | 517          | 2.000e+00                |
| 518          | 0.000e+00        | 518          | -4.263e-14               |
| 519          | 0.000e+00        | 519          | 9.428e-01                |
| 520          | 0.000e+00        | 520          | 1.886e+00                |
| 521          | 0.000e+00        | 521          | 2.087e+00                |
| 522          | 0.000e+00        | 522          | 2.385e+00                |
| 523          | 0.000e+00        | 523          | 2.530e+00                |
| 524          | 0.000e+00        | 524          | 3.795e+00                |
| 525          | 0.000e+00        | 525          | 6.667e-01                |
| 526          | 0.000e+00        | 526          | 1.333e+00                |
| 527          | 0.000e+00        | 527          | 6.667e-01                |
| 528          | 0.000e+00        | 528          | 3.333e+00                |
| 529          | 0.000e+00        | 529          | 2.981e+00                |
| 530          | 0.000e+00        | 530          | 1.491e+00                |
| 531          | 0.000e+00        | 531          | 2.828e+00                |
| 532          | 0.000e+00        | 532          | 0.000e+00                |
| 533          | 0.000e+00        | 533          | 2.000e+00                |
| 534          | 6.667e-01        | 534          | 0.000e+00                |
| 535          | 0.000e+00        | 535          | 2.828e+00                |
| 536          | 0.000e+00        | 536          | 9.570e-14                |
| 537          | 0.000e+00        | 537          | 2.981e+00                |
| 538          | 0.000e+00        | 538          | 1.491e+00                |
| 539          | 1.667e-01        | 539          | 0.000e+00                |
| 540          | 0.000e+00        | 540          | 2.000e+00                |
| 541          | 0.000e+00        | 541          | 2.530e+00                |
| 542          | 0.000e+00        | 542          | 3.795e+00                |
| 543          | 0.000e+00        | 543          | 2.087e+00                |
| 544          | 0.000e+00        | 544          | 2.385e+00                |
| 545          | 0.000e+00        | 545          | 9.428e-01                |
| 546          | 0.000e+00        | 546          | 1.886e+00                |
| 547          | 0.000e+00        | 547          | 2.000e+00                |
| 548          | 0.000e+00        | 548          | 1.421e-14                |
| 549          | 0.000e+00        | 549          | 2.828e+00                |
| 550          | 2.357e-01        | 550          | 0.000e+00                |
| 551          | 0.000e+00        | 551          | 1.779e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 552          | 0.000e+00        | 552          | 6.468e+00                |
| 553          | 0.000e+00        | 553          | 1.265e+00                |
| 554          | 0.000e+00        | 554          | 5.060e+00                |
| 555          | 0.000e+00        | 555          | 0.000e+00                |
| 556          | 0.000e+00        | 556          | 4.472e+00                |
| 557          | 4.318e-16        | 557          | 0.000e+00                |
| 558          | 0.000e+00        | 558          | 2.828e+00                |
| 559          | 0.000e+00        | 559          | 2.000e+00                |
| 560          | 1.000e+00        | 560          | 0.000e+00                |
| 561          | 1.667e-01        | 561          | 0.000e+00                |
| 562          | 0.000e+00        | 562          | 2.000e+00                |
| 563          | 0.000e+00        | 563          | 6.667e-01                |
| 564          | 0.000e+00        | 564          | 3.333e+00                |
| 565          | 0.000e+00        | 565          | 1.333e+00                |
| 566          | 0.000e+00        | 566          | 4.667e+00                |
| 567          | 0.000e+00        | 567          | 1.333e+00                |
| 568          | 0.000e+00        | 568          | 6.667e+00                |
| 569          | 0.000e+00        | 569          | 2.000e+00                |
| 570          | 1.167e+00        | 570          | 0.000e+00                |
| 571          | 0.000e+00        | 571          | -7.327e-15               |
| 572          | 0.000e+00        | 572          | 2.828e+00                |
| 573          | 0.000e+00        | 573          | -1.115e-13               |
| 574          | 0.000e+00        | 574          | 4.472e+00                |
| 575          | 0.000e+00        | 575          | 1.265e+00                |
| 576          | 0.000e+00        | 576          | 5.060e+00                |
| 577          | 0.000e+00        | 577          | 1.779e+00                |
| 578          | 0.000e+00        | 578          | 6.468e+00                |
| 579          | 0.000e+00        | 579          | 6.667e-01                |
| 580          | 0.000e+00        | 580          | 1.333e+00                |
| 581          | 0.000e+00        | 581          | 1.333e+00                |
| 582          | 0.000e+00        | 582          | 2.667e+00                |
| 583          | 0.000e+00        | 583          | 1.333e+00                |
| 584          | 0.000e+00        | 584          | 4.667e+00                |
| 585          | 0.000e+00        | 585          | 2.828e+00                |
| 586          | 2.357e-01        | 586          | 0.000e+00                |
| 587          | 0.000e+00        | 587          | 2.000e+00                |
| 588          | 1.833e+00        | 588          | 0.000e+00                |
| 589          | 0.000e+00        | 589          | 9.428e-01                |
| 590          | 0.000e+00        | 590          | 1.886e+00                |
| 591          | 0.000e+00        | 591          | 2.087e+00                |
| 592          | 0.000e+00        | 592          | 2.385e+00                |
| 593          | 0.000e+00        | 593          | 2.530e+00                |
| 594          | 0.000e+00        | 594          | 3.795e+00                |
| 595          | 0.000e+00        | 595          | 6.667e-01                |
| 596          | 0.000e+00        | 596          | 1.333e+00                |
| 597          | 0.000e+00        | 597          | 6.667e-01                |
| 598          | 0.000e+00        | 598          | 3.333e+00                |
| 599          | 0.000e+00        | 599          | 2.981e+00                |
| 600          | 0.000e+00        | 600          | 1.491e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 601          | 0.000e+00        | 601          | 2.828e+00                |
| 602          | 0.000e+00        | 602          | 3.531e-14                |
| 603          | 0.000e+00        | 603          | 2.000e+00                |
| 604          | 2.667e+00        | 604          | 0.000e+00                |
| 605          | 0.000e+00        | 605          | 2.828e+00                |
| 606          | 0.000e+00        | 606          | 4.241e-14                |
| 607          | 0.000e+00        | 607          | 2.981e+00                |
| 608          | 0.000e+00        | 608          | 1.491e+00                |
| 609          | 0.000e+00        | 609          | 1.421e-14                |
| 610          | 0.000e+00        | 610          | 2.000e+00                |
| 611          | 0.000e+00        | 611          | 2.530e+00                |
| 612          | 0.000e+00        | 612          | 3.795e+00                |
| 613          | 0.000e+00        | 613          | 2.087e+00                |
| 614          | 0.000e+00        | 614          | 2.385e+00                |
| 615          | 0.000e+00        | 615          | 9.428e-01                |
| 616          | 0.000e+00        | 616          | 1.886e+00                |
| 617          | 0.000e+00        | 617          | 2.000e+00                |
| 618          | 2.000e+00        | 618          | 0.000e+00                |
| 619          | 0.000e+00        | 619          | 2.828e+00                |
| 620          | 4.318e-16        | 620          | 0.000e+00                |
| 621          | 0.000e+00        | 621          | 1.779e+00                |
| 622          | 0.000e+00        | 622          | 6.468e+00                |
| 623          | 0.000e+00        | 623          | 1.265e+00                |
| 624          | 0.000e+00        | 624          | 5.060e+00                |
| 625          | -5.536e-32       | 625          | 0.000e+00                |
| 626          | 0.000e+00        | 626          | 4.472e+00                |
| 627          | 2.357e-01        | 627          | 0.000e+00                |
| 628          | 0.000e+00        | 628          | 2.828e+00                |
| 629          | 0.000e+00        | 629          | 2.000e+00                |
| 630          | 1.333e+00        | 630          | 0.000e+00                |
| 631          | 1.667e-01        | 631          | 0.000e+00                |
| 632          | 0.000e+00        | 632          | 2.000e+00                |
| 633          | 0.000e+00        | 633          | 6.667e-01                |
| 634          | 0.000e+00        | 634          | 3.333e+00                |
| 635          | 0.000e+00        | 635          | 1.333e+00                |
| 636          | 0.000e+00        | 636          | 4.667e+00                |
| 637          | 0.000e+00        | 637          | 1.333e+00                |
| 638          | 0.000e+00        | 638          | 6.667e+00                |
| 639          | 0.000e+00        | 639          | 2.000e+00                |
| 640          | 1.333e+00        | 640          | 0.000e+00                |
| 641          | 0.000e+00        | 641          | -2.220e-16               |
| 642          | 0.000e+00        | 642          | 2.828e+00                |
| 643          | 0.000e+00        | 643          | -9.015e-14               |
| 644          | 0.000e+00        | 644          | 4.472e+00                |
| 645          | 0.000e+00        | 645          | 1.265e+00                |
| 646          | 0.000e+00        | 646          | 5.060e+00                |
| 647          | 0.000e+00        | 647          | 1.779e+00                |
| 648          | 0.000e+00        | 648          | 6.468e+00                |
| 649          | 0.000e+00        | 649          | 6.667e-01                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 650          | 0.000e+00        | 650          | 1.333e+00                |
| 651          | 0.000e+00        | 651          | 1.333e+00                |
| 652          | 0.000e+00        | 652          | 2.667e+00                |
| 653          | 0.000e+00        | 653          | 1.333e+00                |
| 654          | 0.000e+00        | 654          | 4.667e+00                |
| 655          | 0.000e+00        | 655          | 2.828e+00                |
| 656          | 2.357e-01        | 656          | 0.000e+00                |
| 657          | 0.000e+00        | 657          | 2.000e+00                |
| 658          | 1.667e+00        | 658          | 0.000e+00                |
| 659          | 0.000e+00        | 659          | 9.428e-01                |
| 660          | 0.000e+00        | 660          | 1.886e+00                |
| 661          | 0.000e+00        | 661          | 2.087e+00                |
| 662          | 0.000e+00        | 662          | 2.385e+00                |
| 663          | 0.000e+00        | 663          | 2.530e+00                |
| 664          | 0.000e+00        | 664          | 3.795e+00                |
| 665          | 0.000e+00        | 665          | 6.667e-01                |
| 666          | 0.000e+00        | 666          | 1.333e+00                |
| 667          | 0.000e+00        | 667          | 6.667e-01                |
| 668          | 0.000e+00        | 668          | 3.333e+00                |
| 669          | 0.000e+00        | 669          | 2.981e+00                |
| 670          | 0.000e+00        | 670          | 1.491e+00                |
| 671          | 0.000e+00        | 671          | 2.828e+00                |
| 672          | 0.000e+00        | 672          | 1.044e-14                |
| 673          | 0.000e+00        | 673          | 2.000e+00                |
| 674          | 2.667e+00        | 674          | 0.000e+00                |
| 675          | 0.000e+00        | 675          | 2.828e+00                |
| 676          | 0.000e+00        | 676          | 3.531e-14                |
| 677          | 0.000e+00        | 677          | 2.981e+00                |
| 678          | 0.000e+00        | 678          | 1.491e+00                |
| 679          | 0.000e+00        | 679          | -1.421e-14               |
| 680          | 0.000e+00        | 680          | 2.000e+00                |
| 681          | 0.000e+00        | 681          | 2.530e+00                |
| 682          | 0.000e+00        | 682          | 3.795e+00                |
| 683          | 0.000e+00        | 683          | 2.087e+00                |
| 684          | 0.000e+00        | 684          | 2.385e+00                |
| 685          | 0.000e+00        | 685          | 9.428e-01                |
| 686          | 0.000e+00        | 686          | 1.886e+00                |
| 687          | 0.000e+00        | 687          | 2.000e+00                |
| 688          | 1.667e+00        | 688          | 0.000e+00                |
| 689          | 0.000e+00        | 689          | 2.828e+00                |
| 690          | 2.357e-01        | 690          | 0.000e+00                |
| 691          | 0.000e+00        | 691          | 1.779e+00                |
| 692          | 0.000e+00        | 692          | 6.468e+00                |
| 693          | 0.000e+00        | 693          | 1.265e+00                |
| 694          | 0.000e+00        | 694          | 5.060e+00                |
| 695          | 3.413e-16        | 695          | 0.000e+00                |
| 696          | 0.000e+00        | 696          | 4.472e+00                |
| 697          | 0.000e+00        | 697          | -3.575e-14               |
| 698          | 0.000e+00        | 698          | 2.828e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 699          | 0.000e+00        | 699          | 2.000e+00                |
| 700          | 1.333e+00        | 700          | 0.000e+00                |
| 701          | 1.667e-01        | 701          | 0.000e+00                |
| 702          | 0.000e+00        | 702          | 2.000e+00                |
| 703          | 0.000e+00        | 703          | 6.667e-01                |
| 704          | 0.000e+00        | 704          | 3.333e+00                |
| 705          | 0.000e+00        | 705          | 1.333e+00                |
| 706          | 0.000e+00        | 706          | 4.667e+00                |
| 707          | 0.000e+00        | 707          | 1.333e+00                |
| 708          | 0.000e+00        | 708          | 6.667e+00                |
| 709          | 0.000e+00        | 709          | 2.000e+00                |
| 710          | 1.500e+00        | 710          | 0.000e+00                |
| 711          | 0.000e+00        | 711          | -2.220e-16               |
| 712          | 0.000e+00        | 712          | 2.828e+00                |
| 713          | 0.000e+00        | 713          | 0.000e+00                |
| 714          | 0.000e+00        | 714          | 4.472e+00                |
| 715          | 0.000e+00        | 715          | 1.265e+00                |
| 716          | 0.000e+00        | 716          | 5.060e+00                |
| 717          | 0.000e+00        | 717          | 1.779e+00                |
| 718          | 0.000e+00        | 718          | 6.468e+00                |
| 719          | 0.000e+00        | 719          | 6.667e-01                |
| 720          | 0.000e+00        | 720          | 1.333e+00                |
| 721          | 0.000e+00        | 721          | 1.333e+00                |
| 722          | 0.000e+00        | 722          | 2.667e+00                |
| 723          | 0.000e+00        | 723          | 1.333e+00                |
| 724          | 0.000e+00        | 724          | 4.667e+00                |
| 725          | 0.000e+00        | 725          | 2.828e+00                |
| 726          | 2.357e-01        | 726          | 0.000e+00                |
| 727          | 0.000e+00        | 727          | 2.000e+00                |
| 728          | 1.500e+00        | 728          | 0.000e+00                |
| 729          | 0.000e+00        | 729          | 9.428e-01                |
| 730          | 0.000e+00        | 730          | 1.886e+00                |
| 731          | 0.000e+00        | 731          | 2.087e+00                |
| 732          | 0.000e+00        | 732          | 2.385e+00                |
| 733          | 0.000e+00        | 733          | 2.530e+00                |
| 734          | 0.000e+00        | 734          | 3.795e+00                |
| 735          | 0.000e+00        | 735          | 6.667e-01                |
| 736          | 0.000e+00        | 736          | 1.333e+00                |
| 737          | 0.000e+00        | 737          | 6.667e-01                |
| 738          | 0.000e+00        | 738          | 3.333e+00                |
| 739          | 0.000e+00        | 739          | 2.981e+00                |
| 740          | 0.000e+00        | 740          | 1.491e+00                |
| 741          | 0.000e+00        | 741          | 2.828e+00                |
| 742          | 5.238e-01        | 742          | 0.000e+00                |
| 743          | 0.000e+00        | 743          | 2.000e+00                |
| 744          | 1.926e+00        | 744          | 0.000e+00                |
| 745          | 0.000e+00        | 745          | 2.828e+00                |
| 746          | 5.238e-01        | 746          | 0.000e+00                |
| 747          | 0.000e+00        | 747          | 2.981e+00                |

Continua na próxima página



| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 748          | 0.000e+00        | 748          | 1.491e+00                |
| 749          | 0.000e+00        | 749          | 5.773e-15                |
| 750          | 0.000e+00        | 750          | 2.000e+00                |
| 751          | 0.000e+00        | 751          | 2.530e+00                |
| 752          | 0.000e+00        | 752          | 3.795e+00                |
| 753          | 0.000e+00        | 753          | 2.087e+00                |
| 754          | 0.000e+00        | 754          | 2.385e+00                |
| 755          | 0.000e+00        | 755          | 9.428e-01                |
| 756          | 0.000e+00        | 756          | 1.886e+00                |
| 757          | 0.000e+00        | 757          | 2.000e+00                |
| 758          | 1.667e+00        | 758          | 0.000e+00                |
| 759          | 0.000e+00        | 759          | 2.828e+00                |
| 760          | 7.850e-16        | 760          | 0.000e+00                |
| 761          | 0.000e+00        | 761          | 1.779e+00                |
| 762          | 0.000e+00        | 762          | 6.468e+00                |
| 763          | 0.000e+00        | 763          | 1.265e+00                |
| 764          | 0.000e+00        | 764          | 5.060e+00                |
| 765          | 0.000e+00        | 765          | 2.887e-14                |
| 766          | 0.000e+00        | 766          | 4.472e+00                |
| 767          | 2.357e-01        | 767          | 0.000e+00                |
| 768          | 0.000e+00        | 768          | 2.828e+00                |
| 769          | 0.000e+00        | 769          | 2.000e+00                |
| 770          | 1.667e+00        | 770          | 0.000e+00                |
| 771          | 1.667e-01        | 771          | 0.000e+00                |
| 772          | 0.000e+00        | 772          | 2.000e+00                |
| 773          | 0.000e+00        | 773          | 6.667e-01                |
| 774          | 0.000e+00        | 774          | 3.333e+00                |
| 775          | 0.000e+00        | 775          | 1.333e+00                |
| 776          | 0.000e+00        | 776          | 4.667e+00                |
| 777          | 0.000e+00        | 777          | 1.333e+00                |
| 778          | 0.000e+00        | 778          | 6.667e+00                |
| 779          | 0.000e+00        | 779          | 2.000e+00                |
| 780          | 1.667e+00        | 780          | 0.000e+00                |
| 781          | 0.000e+00        | 781          | 3.331e-15                |
| 782          | 0.000e+00        | 782          | 2.828e+00                |
| 783          | 3.650e-16        | 783          | 0.000e+00                |
| 784          | 0.000e+00        | 784          | 4.472e+00                |
| 785          | 0.000e+00        | 785          | 1.265e+00                |
| 786          | 0.000e+00        | 786          | 5.060e+00                |
| 787          | 0.000e+00        | 787          | 1.779e+00                |
| 788          | 0.000e+00        | 788          | 6.468e+00                |
| 789          | 0.000e+00        | 789          | 6.667e-01                |
| 790          | 0.000e+00        | 790          | 1.333e+00                |
| 791          | 0.000e+00        | 791          | 1.333e+00                |
| 792          | 0.000e+00        | 792          | 2.667e+00                |
| 793          | 0.000e+00        | 793          | 1.333e+00                |
| 794          | 0.000e+00        | 794          | 4.667e+00                |
| 795          | 0.000e+00        | 795          | 2.828e+00                |
| 796          | 7.595e-01        | 796          | 0.000e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 797          | 0.000e+00        | 797          | 2.000e+00                |
| 798          | 1.333e+00        | 798          | 0.000e+00                |
| 799          | 0.000e+00        | 799          | 9.428e-01                |
| 800          | 0.000e+00        | 800          | 1.886e+00                |
| 801          | 0.000e+00        | 801          | 2.087e+00                |
| 802          | 0.000e+00        | 802          | 2.385e+00                |
| 803          | 0.000e+00        | 803          | 2.530e+00                |
| 804          | 0.000e+00        | 804          | 3.795e+00                |
| 805          | 0.000e+00        | 805          | 6.667e-01                |
| 806          | 0.000e+00        | 806          | 1.333e+00                |
| 807          | 0.000e+00        | 807          | 6.667e-01                |
| 808          | 0.000e+00        | 808          | 3.333e+00                |
| 809          | 0.000e+00        | 809          | 2.981e+00                |
| 810          | 0.000e+00        | 810          | 1.491e+00                |
| 811          | 0.000e+00        | 811          | 2.828e+00                |
| 812          | 0.000e+00        | 812          | -7.327e-15               |
| 813          | 0.000e+00        | 813          | 2.000e+00                |
| 814          | 1.926e+00        | 814          | 0.000e+00                |
| 815          | 0.000e+00        | 815          | 2.828e+00                |
| 816          | 0.000e+00        | 816          | 3.175e-14                |
| 817          | 0.000e+00        | 817          | 2.981e+00                |
| 818          | 0.000e+00        | 818          | 1.491e+00                |
| 819          | 0.000e+00        | 819          | -1.465e-14               |
| 820          | 0.000e+00        | 820          | 2.000e+00                |
| 821          | 0.000e+00        | 821          | 2.530e+00                |
| 822          | 0.000e+00        | 822          | 3.795e+00                |
| 823          | 0.000e+00        | 823          | 2.087e+00                |
| 824          | 0.000e+00        | 824          | 2.385e+00                |
| 825          | 0.000e+00        | 825          | 9.428e-01                |
| 826          | 0.000e+00        | 826          | 1.886e+00                |
| 827          | 0.000e+00        | 827          | 2.000e+00                |
| 828          | 1.333e+00        | 828          | 0.000e+00                |
| 829          | 0.000e+00        | 829          | 2.828e+00                |
| 830          | 7.595e-01        | 830          | 0.000e+00                |
| 831          | 0.000e+00        | 831          | 1.779e+00                |
| 832          | 0.000e+00        | 832          | 6.468e+00                |
| 833          | 0.000e+00        | 833          | 1.265e+00                |
| 834          | 0.000e+00        | 834          | 5.060e+00                |
| 835          | 3.650e-16        | 835          | 0.000e+00                |
| 836          | 0.000e+00        | 836          | 4.472e+00                |
| 837          | 0.000e+00        | 837          | -1.799e-14               |
| 838          | 0.000e+00        | 838          | 2.828e+00                |
| 839          | 0.000e+00        | 839          | 2.000e+00                |
| 840          | 1.667e+00        | 840          | 0.000e+00                |
| 841          | 5.370e-01        | 841          | 0.000e+00                |
| 842          | 0.000e+00        | 842          | 2.000e+00                |
| 843          | 0.000e+00        | 843          | 6.667e-01                |
| 844          | 0.000e+00        | 844          | 3.333e+00                |
| 845          | 0.000e+00        | 845          | 1.333e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 846          | 0.000e+00        | 846          | 4.667e+00                |
| 847          | 0.000e+00        | 847          | 1.333e+00                |
| 848          | 0.000e+00        | 848          | 6.667e+00                |
| 849          | 0.000e+00        | 849          | 2.000e+00                |
| 850          | 2.204e+00        | 850          | 0.000e+00                |
| 851          | 0.000e+00        | 851          | 6.883e-15                |
| 852          | 0.000e+00        | 852          | 2.828e+00                |
| 853          | 0.000e+00        | 853          | 0.000e+00                |
| 854          | 0.000e+00        | 854          | 4.472e+00                |
| 855          | 0.000e+00        | 855          | 1.265e+00                |
| 856          | 0.000e+00        | 856          | 5.060e+00                |
| 857          | 0.000e+00        | 857          | 1.779e+00                |
| 858          | 0.000e+00        | 858          | 6.468e+00                |
| 859          | 0.000e+00        | 859          | 6.667e-01                |
| 860          | 0.000e+00        | 860          | 1.333e+00                |
| 861          | 0.000e+00        | 861          | 1.333e+00                |
| 862          | 0.000e+00        | 862          | 2.667e+00                |
| 863          | 0.000e+00        | 863          | 1.333e+00                |
| 864          | 0.000e+00        | 864          | 4.667e+00                |
| 865          | 0.000e+00        | 865          | 2.828e+00                |
| 866          | 7.595e-01        | 866          | 0.000e+00                |
| 867          | 0.000e+00        | 867          | 2.000e+00                |
| 868          | 7.963e-01        | 868          | 0.000e+00                |
| 869          | 0.000e+00        | 869          | 9.428e-01                |
| 870          | 0.000e+00        | 870          | 1.886e+00                |
| 871          | 0.000e+00        | 871          | 2.087e+00                |
| 872          | 0.000e+00        | 872          | 2.385e+00                |
| 873          | 0.000e+00        | 873          | 2.530e+00                |
| 874          | 0.000e+00        | 874          | 3.795e+00                |
| 875          | 0.000e+00        | 875          | 6.667e-01                |
| 876          | 0.000e+00        | 876          | 1.333e+00                |
| 877          | 0.000e+00        | 877          | 6.667e-01                |
| 878          | 0.000e+00        | 878          | 3.333e+00                |
| 879          | 0.000e+00        | 879          | 2.981e+00                |
| 880          | 0.000e+00        | 880          | 1.491e+00                |
| 881          | 0.000e+00        | 881          | 2.828e+00                |
| 882          | 0.000e+00        | 882          | -6.772e-14               |
| 883          | 0.000e+00        | 883          | 2.000e+00                |
| 884          | 1.926e+00        | 884          | 0.000e+00                |
| 885          | 0.000e+00        | 885          | 2.828e+00                |
| 886          | 0.000e+00        | 886          | 9.570e-14                |
| 887          | 0.000e+00        | 887          | 2.981e+00                |
| 888          | 0.000e+00        | 888          | 1.491e+00                |
| 889          | 5.370e-01        | 889          | 0.000e+00                |
| 890          | 0.000e+00        | 890          | 2.000e+00                |
| 891          | 0.000e+00        | 891          | 2.530e+00                |
| 892          | 0.000e+00        | 892          | 3.795e+00                |
| 893          | 0.000e+00        | 893          | 2.087e+00                |
| 894          | 0.000e+00        | 894          | 2.385e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 895          | 0.000e+00        | 895          | 9.428e-01                |
| 896          | 0.000e+00        | 896          | 1.886e+00                |
| 897          | 0.000e+00        | 897          | 2.000e+00                |
| 898          | 7.963e-01        | 898          | 0.000e+00                |
| 899          | 0.000e+00        | 899          | 2.828e+00                |
| 900          | 7.595e-01        | 900          | 0.000e+00                |
| 901          | 0.000e+00        | 901          | 1.779e+00                |
| 902          | 0.000e+00        | 902          | 6.468e+00                |
| 903          | 0.000e+00        | 903          | 1.265e+00                |
| 904          | 0.000e+00        | 904          | 5.060e+00                |
| 905          | 0.000e+00        | 905          | -4.219e-14               |
| 906          | 0.000e+00        | 906          | 4.472e+00                |
| 907          | 0.000e+00        | 907          | -2.220e-16               |
| 908          | 0.000e+00        | 908          | 2.828e+00                |
| 909          | 0.000e+00        | 909          | 2.000e+00                |
| 910          | 2.204e+00        | 910          | 0.000e+00                |
| 911          | 5.370e-01        | 911          | 0.000e+00                |
| 912          | 0.000e+00        | 912          | 2.000e+00                |
| 913          | 0.000e+00        | 913          | 6.667e-01                |
| 914          | 0.000e+00        | 914          | 3.333e+00                |
| 915          | 0.000e+00        | 915          | 1.333e+00                |
| 916          | 0.000e+00        | 916          | 4.667e+00                |
| 917          | 0.000e+00        | 917          | 1.333e+00                |
| 918          | 0.000e+00        | 918          | 6.667e+00                |
| 919          | 0.000e+00        | 919          | 2.000e+00                |
| 920          | 2.741e+00        | 920          | 0.000e+00                |
| 921          | 0.000e+00        | 921          | -7.327e-15               |
| 922          | 0.000e+00        | 922          | 2.828e+00                |
| 923          | 0.000e+00        | 923          | -1.910e-14               |
| 924          | 0.000e+00        | 924          | 4.472e+00                |
| 925          | 0.000e+00        | 925          | 1.265e+00                |
| 926          | 0.000e+00        | 926          | 5.060e+00                |
| 927          | 0.000e+00        | 927          | 1.779e+00                |
| 928          | 0.000e+00        | 928          | 6.468e+00                |
| 929          | 0.000e+00        | 929          | 6.667e-01                |
| 930          | 0.000e+00        | 930          | 1.333e+00                |
| 931          | 0.000e+00        | 931          | 1.333e+00                |
| 932          | 0.000e+00        | 932          | 2.667e+00                |
| 933          | 0.000e+00        | 933          | 1.333e+00                |
| 934          | 0.000e+00        | 934          | 4.667e+00                |
| 935          | 0.000e+00        | 935          | 2.828e+00                |
| 936          | 7.595e-01        | 936          | 0.000e+00                |
| 937          | 0.000e+00        | 937          | 2.000e+00                |
| 938          | 2.593e-01        | 938          | 0.000e+00                |
| 939          | 0.000e+00        | 939          | 9.428e-01                |
| 940          | 0.000e+00        | 940          | 1.886e+00                |
| 941          | 0.000e+00        | 941          | 2.087e+00                |
| 942          | 0.000e+00        | 942          | 2.385e+00                |
| 943          | 0.000e+00        | 943          | 2.530e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 944          | 0.000e+00        | 944          | 3.795e+00                |
| 945          | 0.000e+00        | 945          | 6.667e-01                |
| 946          | 0.000e+00        | 946          | 1.333e+00                |
| 947          | 0.000e+00        | 947          | 6.667e-01                |
| 948          | 0.000e+00        | 948          | 3.333e+00                |
| 949          | 0.000e+00        | 949          | 2.981e+00                |
| 950          | 0.000e+00        | 950          | 1.491e+00                |
| 951          | 0.000e+00        | 951          | 2.828e+00                |
| 952          | 0.000e+00        | 952          | -8.193e-14               |
| 953          | 0.000e+00        | 953          | 2.000e+00                |
| 954          | 1.926e+00        | 954          | 0.000e+00                |
| 955          | 0.000e+00        | 955          | 2.828e+00                |
| 956          | 0.000e+00        | 956          | 8.149e-14                |
| 957          | 0.000e+00        | 957          | 2.981e+00                |
| 958          | 0.000e+00        | 958          | 1.491e+00                |
| 959          | 5.370e-01        | 959          | 0.000e+00                |
| 960          | 0.000e+00        | 960          | 2.000e+00                |
| 961          | 0.000e+00        | 961          | 2.530e+00                |
| 962          | 0.000e+00        | 962          | 3.795e+00                |
| 963          | 0.000e+00        | 963          | 2.087e+00                |
| 964          | 0.000e+00        | 964          | 2.385e+00                |
| 965          | 0.000e+00        | 965          | 9.428e-01                |
| 966          | 0.000e+00        | 966          | 1.886e+00                |
| 967          | 0.000e+00        | 967          | 2.000e+00                |
| 968          | 2.593e-01        | 968          | 0.000e+00                |
| 969          | 0.000e+00        | 969          | 2.828e+00                |
| 970          | 7.595e-01        | 970          | 0.000e+00                |
| 971          | 0.000e+00        | 971          | 1.779e+00                |
| 972          | 0.000e+00        | 972          | 6.468e+00                |
| 973          | 0.000e+00        | 973          | 1.265e+00                |
| 974          | 0.000e+00        | 974          | 5.060e+00                |
| 975          | 0.000e+00        | 975          | 0.000e+00                |
| 976          | 0.000e+00        | 976          | 4.472e+00                |
| 977          | 0.000e+00        | 977          | -3.775e-15               |
| 978          | 0.000e+00        | 978          | 2.828e+00                |
| 979          | 0.000e+00        | 979          | 2.000e+00                |
| 980          | 2.741e+00        | 980          | 0.000e+00                |
| 981          | 2.593e-01        | 981          | 0.000e+00                |
| 982          | 0.000e+00        | 982          | 2.000e+00                |
| 983          | 0.000e+00        | 983          | 6.667e-01                |
| 984          | 0.000e+00        | 984          | 3.333e+00                |
| 985          | 0.000e+00        | 985          | 1.333e+00                |
| 986          | 0.000e+00        | 986          | 4.667e+00                |
| 987          | 0.000e+00        | 987          | 1.333e+00                |
| 988          | 0.000e+00        | 988          | 6.667e+00                |
| 989          | 0.000e+00        | 989          | 2.000e+00                |
| 990          | 3.417e+00        | 990          | 0.000e+00                |
| 991          | 0.000e+00        | 991          | -2.220e-16               |
| 992          | 0.000e+00        | 992          | 2.828e+00                |

Continua na próxima página

| Coordenada x | Valor x (Primal) | Coordenada z | Valor z (Custo Reduzido) |
|--------------|------------------|--------------|--------------------------|
| 993          | 3.106e-01        | 993          | 0.000e+00                |
| 994          | 0.000e+00        | 994          | 4.472e+00                |
| 995          | 0.000e+00        | 995          | 1.265e+00                |
| 996          | 0.000e+00        | 996          | 5.060e+00                |
| 997          | 0.000e+00        | 997          | 1.779e+00                |
| 998          | 0.000e+00        | 998          | 6.468e+00                |
| 999          | 0.000e+00        | 999          | 6.667e-01                |
| 1000         | 0.000e+00        | 1000         | 1.333e+00                |

### 3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema scsd8

| Coordenada y | Valor y (Dual) |
|--------------|----------------|
| 1            | 3.333e+00      |
| 2            | -3.900e+01     |
| 3            | 2.333e+00      |
| 4            | -4.100e+01     |
| 5            | 1.333e+00      |
| 6            | -4.233e+01     |
| 7            | 3.333e-01      |
| 8            | -4.100e+01     |
| 9            | -6.667e-01     |
| 10           | -3.900e+01     |
| 11           | 1.333e+00      |
| 12           | -3.800e+01     |
| 13           | 1.000e+00      |
| 14           | -4.000e+01     |
| 15           | 1.333e+00      |
| 16           | -4.133e+01     |
| 17           | 1.667e+00      |
| 18           | -4.000e+01     |
| 19           | 1.333e+00      |
| 20           | -3.800e+01     |
| 21           | -3.837e-13     |
| 22           | -3.700e+01     |
| 23           | 1.000e+00      |
| 24           | -3.900e+01     |
| 25           | 1.333e+00      |
| 26           | -4.033e+01     |
| 27           | 1.667e+00      |
| 28           | -3.900e+01     |
| 29           | 2.667e+00      |
| 30           | -3.700e+01     |
| 31           | -3.908e-13     |
| 32           | -3.600e+01     |

Continua na próxima página

| Coordenada y | Valor y (Dual) |
|--------------|----------------|
| 33           | 1.000e+00      |
| 34           | -3.800e+01     |
| 35           | 1.333e+00      |
| 36           | -3.933e+01     |
| 37           | 1.667e+00      |
| 38           | -3.800e+01     |
| 39           | 2.667e+00      |
| 40           | -3.600e+01     |
| 41           | -3.949e-13     |
| 42           | -3.500e+01     |
| 43           | 1.000e+00      |
| 44           | -3.700e+01     |
| 45           | 1.333e+00      |
| 46           | -3.833e+01     |
| 47           | 1.667e+00      |
| 48           | -3.700e+01     |
| 49           | 2.667e+00      |
| 50           | -3.500e+01     |
| 51           | -4.263e-13     |
| 52           | -3.400e+01     |
| 53           | 1.000e+00      |
| 54           | -3.600e+01     |
| 55           | 1.333e+00      |
| 56           | -3.733e+01     |
| 57           | 1.667e+00      |
| 58           | -3.600e+01     |
| 59           | 2.667e+00      |
| 60           | -3.400e+01     |
| 61           | -4.405e-13     |
| 62           | -3.300e+01     |
| 63           | 1.000e+00      |
| 64           | -3.500e+01     |
| 65           | 1.333e+00      |
| 66           | -3.633e+01     |
| 67           | 1.667e+00      |
| 68           | -3.500e+01     |
| 69           | 2.667e+00      |
| 70           | -3.300e+01     |
| 71           | -4.476e-13     |
| 72           | -3.200e+01     |
| 73           | 1.000e+00      |
| 74           | -3.400e+01     |
| 75           | 1.333e+00      |
| 76           | -3.533e+01     |
| 77           | 1.667e+00      |
| 78           | -3.400e+01     |
| 79           | 2.667e+00      |
| 80           | -3.200e+01     |
| 81           | -4.547e-13     |

Continua na próxima página

| Coordenada y               | Valor y (Dual) |
|----------------------------|----------------|
| 82                         | -3.100e+01     |
| 83                         | 1.000e+00      |
| 84                         | -3.300e+01     |
| 85                         | 1.333e+00      |
| 86                         | -3.433e+01     |
| 87                         | 1.667e+00      |
| 88                         | -3.300e+01     |
| 89                         | 2.667e+00      |
| 90                         | -3.100e+01     |
| 91                         | -4.121e-13     |
| 92                         | -3.000e+01     |
| 93                         | 1.000e+00      |
| 94                         | -3.200e+01     |
| 95                         | 1.333e+00      |
| 96                         | -3.333e+01     |
| 97                         | 1.667e+00      |
| 98                         | -3.200e+01     |
| 99                         | 2.667e+00      |
| 100                        | -3.000e+01     |
| 101                        | -3.766e-13     |
| 102                        | -2.900e+01     |
| 103                        | 1.000e+00      |
| 104                        | -3.100e+01     |
| 105                        | 1.333e+00      |
| 106                        | -3.233e+01     |
| 107                        | 1.667e+00      |
| 108                        | -3.100e+01     |
| 109                        | 2.667e+00      |
| 110                        | -2.900e+01     |
| 111                        | -3.411e-13     |
| 112                        | -2.800e+01     |
| 113                        | 1.000e+00      |
| 114                        | -3.000e+01     |
| 115                        | 1.333e+00      |
| 116                        | -3.133e+01     |
| 117                        | 1.667e+00      |
| 118                        | -3.000e+01     |
| 119                        | 2.667e+00      |
| 120                        | -2.800e+01     |
| 121                        | -3.126e-13     |
| 122                        | -2.700e+01     |
| 123                        | 1.000e+00      |
| 124                        | -2.900e+01     |
| 125                        | 1.333e+00      |
| 126                        | -3.033e+01     |
| 127                        | 1.667e+00      |
| 128                        | -2.900e+01     |
| 129                        | 2.667e+00      |
| 130                        | -2.700e+01     |
| Continua na próxima página |                |



| Coordenada y               | Valor y (Dual) |
|----------------------------|----------------|
| 131                        | -2.842e-13     |
| 132                        | -2.600e+01     |
| 133                        | 1.000e+00      |
| 134                        | -2.800e+01     |
| 135                        | 1.333e+00      |
| 136                        | -2.933e+01     |
| 137                        | 1.667e+00      |
| 138                        | -2.800e+01     |
| 139                        | 2.667e+00      |
| 140                        | -2.600e+01     |
| 141                        | -2.416e-13     |
| 142                        | -2.500e+01     |
| 143                        | 1.000e+00      |
| 144                        | -2.700e+01     |
| 145                        | 1.333e+00      |
| 146                        | -2.833e+01     |
| 147                        | 1.667e+00      |
| 148                        | -2.700e+01     |
| 149                        | 2.667e+00      |
| 150                        | -2.500e+01     |
| 151                        | -2.061e-13     |
| 152                        | -2.400e+01     |
| 153                        | 1.000e+00      |
| 154                        | -2.600e+01     |
| 155                        | 1.333e+00      |
| 156                        | -2.733e+01     |
| 157                        | 1.667e+00      |
| 158                        | -2.600e+01     |
| 159                        | 2.667e+00      |
| 160                        | -2.400e+01     |
| 161                        | -1.705e-13     |
| 162                        | -2.300e+01     |
| 163                        | 1.000e+00      |
| 164                        | -2.500e+01     |
| 165                        | 1.333e+00      |
| 166                        | -2.633e+01     |
| 167                        | 1.667e+00      |
| 168                        | -2.500e+01     |
| 169                        | 2.667e+00      |
| 170                        | -2.300e+01     |
| 171                        | -1.510e-13     |
| 172                        | -2.200e+01     |
| 173                        | 1.000e+00      |
| 174                        | -2.400e+01     |
| 175                        | 1.333e+00      |
| 176                        | -2.533e+01     |
| 177                        | 1.667e+00      |
| 178                        | -2.400e+01     |
| 179                        | 2.667e+00      |
| Continua na próxima página |                |

| Coordenada y               | Valor y (Dual) |
|----------------------------|----------------|
| 180                        | -2.200e+01     |
| 181                        | -1.990e-13     |
| 182                        | -2.100e+01     |
| 183                        | 1.000e+00      |
| 184                        | -2.300e+01     |
| 185                        | 1.333e+00      |
| 186                        | -2.433e+01     |
| 187                        | 1.667e+00      |
| 188                        | -2.300e+01     |
| 189                        | 2.667e+00      |
| 190                        | -2.100e+01     |
| 191                        | -1.972e-13     |
| 192                        | -2.000e+01     |
| 193                        | 1.000e+00      |
| 194                        | -2.200e+01     |
| 195                        | 1.333e+00      |
| 196                        | -2.333e+01     |
| 197                        | 1.667e+00      |
| 198                        | -2.200e+01     |
| 199                        | 2.667e+00      |
| 200                        | -2.000e+01     |
| 201                        | -1.634e-13     |
| 202                        | -1.900e+01     |
| 203                        | 1.000e+00      |
| 204                        | -2.100e+01     |
| 205                        | 1.333e+00      |
| 206                        | -2.233e+01     |
| 207                        | 1.667e+00      |
| 208                        | -2.100e+01     |
| 209                        | 2.667e+00      |
| 210                        | -1.900e+01     |
| 211                        | -1.948e-13     |
| 212                        | -1.800e+01     |
| 213                        | 1.000e+00      |
| 214                        | -2.000e+01     |
| 215                        | 1.333e+00      |
| 216                        | -2.133e+01     |
| 217                        | 1.667e+00      |
| 218                        | -2.000e+01     |
| 219                        | 2.667e+00      |
| 220                        | -1.800e+01     |
| 221                        | -2.451e-13     |
| 222                        | -1.700e+01     |
| 223                        | 1.000e+00      |
| 224                        | -1.900e+01     |
| 225                        | 1.333e+00      |
| 226                        | -2.033e+01     |
| 227                        | 1.667e+00      |
| 228                        | -1.900e+01     |
| Continua na próxima página |                |

| Coordenada y | Valor y (Dual) |
|--------------|----------------|
| 229          | 2.667e+00      |
| 230          | -1.700e+01     |
| 231          | -9.237e-14     |
| 232          | -1.600e+01     |
| 233          | 1.000e+00      |
| 234          | -1.800e+01     |
| 235          | 1.333e+00      |
| 236          | -1.933e+01     |
| 237          | 1.667e+00      |
| 238          | -1.800e+01     |
| 239          | 2.667e+00      |
| 240          | -1.600e+01     |
| 241          | -8.527e-14     |
| 242          | -1.500e+01     |
| 243          | 1.000e+00      |
| 244          | -1.700e+01     |
| 245          | 1.333e+00      |
| 246          | -1.833e+01     |
| 247          | 1.667e+00      |
| 248          | -1.700e+01     |
| 249          | 2.667e+00      |
| 250          | -1.500e+01     |
| 251          | -7.816e-14     |
| 252          | -1.400e+01     |
| 253          | 1.000e+00      |
| 254          | -1.600e+01     |
| 255          | 1.333e+00      |
| 256          | -1.733e+01     |
| 257          | 1.667e+00      |
| 258          | -1.600e+01     |
| 259          | 2.667e+00      |
| 260          | -1.400e+01     |
| 261          | -7.105e-14     |
| 262          | -1.300e+01     |
| 263          | 1.000e+00      |
| 264          | -1.500e+01     |
| 265          | 1.333e+00      |
| 266          | -1.633e+01     |
| 267          | 1.667e+00      |
| 268          | -1.500e+01     |
| 269          | 2.667e+00      |
| 270          | -1.300e+01     |
| 271          | -6.395e-14     |
| 272          | -1.200e+01     |
| 273          | 1.000e+00      |
| 274          | -1.400e+01     |
| 275          | 1.333e+00      |
| 276          | -1.533e+01     |
| 277          | 1.667e+00      |

Continua na próxima página

| Coordenada y               | Valor y (Dual) |
|----------------------------|----------------|
| 278                        | -1.400e+01     |
| 279                        | 2.667e+00      |
| 280                        | -1.200e+01     |
| 281                        | -5.684e-14     |
| 282                        | -1.100e+01     |
| 283                        | 1.000e+00      |
| 284                        | -1.300e+01     |
| 285                        | 1.333e+00      |
| 286                        | -1.433e+01     |
| 287                        | 1.667e+00      |
| 288                        | -1.300e+01     |
| 289                        | 2.667e+00      |
| 290                        | -1.100e+01     |
| 291                        | -4.974e-14     |
| 292                        | -1.000e+01     |
| 293                        | 1.000e+00      |
| 294                        | -1.200e+01     |
| 295                        | 1.333e+00      |
| 296                        | -1.333e+01     |
| 297                        | 1.667e+00      |
| 298                        | -1.200e+01     |
| 299                        | 2.667e+00      |
| 300                        | -1.000e+01     |
| 301                        | -4.263e-14     |
| 302                        | -9.000e+00     |
| 303                        | 1.000e+00      |
| 304                        | -1.100e+01     |
| 305                        | 1.333e+00      |
| 306                        | -1.233e+01     |
| 307                        | 1.667e+00      |
| 308                        | -1.100e+01     |
| 309                        | 2.667e+00      |
| 310                        | -9.000e+00     |
| 311                        | -3.553e-14     |
| 312                        | -8.000e+00     |
| 313                        | 1.000e+00      |
| 314                        | -1.000e+01     |
| 315                        | 1.333e+00      |
| 316                        | -1.133e+01     |
| 317                        | 1.667e+00      |
| 318                        | -1.000e+01     |
| 319                        | 2.667e+00      |
| 320                        | -8.000e+00     |
| 321                        | -4.263e-14     |
| 322                        | -7.000e+00     |
| 323                        | 1.000e+00      |
| 324                        | -9.000e+00     |
| 325                        | 1.333e+00      |
| 326                        | -1.033e+01     |
| Continua na próxima página |                |

| Coordenada y | Valor y (Dual) |
|--------------|----------------|
| 327          | 1.667e+00      |
| 328          | -9.000e+00     |
| 329          | 2.667e+00      |
| 330          | -7.000e+00     |
| 331          | -3.730e-14     |
| 332          | -6.000e+00     |
| 333          | 1.000e+00      |
| 334          | -8.000e+00     |
| 335          | 1.333e+00      |
| 336          | -9.333e+00     |
| 337          | 1.667e+00      |
| 338          | -8.000e+00     |
| 339          | 2.667e+00      |
| 340          | -6.000e+00     |
| 341          | -3.197e-14     |
| 342          | -5.000e+00     |
| 343          | 1.000e+00      |
| 344          | -7.000e+00     |
| 345          | 1.333e+00      |
| 346          | -8.333e+00     |
| 347          | 1.667e+00      |
| 348          | -7.000e+00     |
| 349          | 2.667e+00      |
| 350          | -5.000e+00     |
| 351          | -3.197e-14     |
| 352          | -4.000e+00     |
| 353          | 1.000e+00      |
| 354          | -6.000e+00     |
| 355          | 1.333e+00      |
| 356          | -7.333e+00     |
| 357          | 1.667e+00      |
| 358          | -6.000e+00     |
| 359          | 2.667e+00      |
| 360          | -4.000e+00     |
| 361          | -2.665e-14     |
| 362          | -3.000e+00     |
| 363          | 1.000e+00      |
| 364          | -5.000e+00     |
| 365          | 1.333e+00      |
| 366          | -6.333e+00     |
| 367          | 1.667e+00      |
| 368          | -5.000e+00     |
| 369          | 2.667e+00      |
| 370          | -3.000e+00     |
| 371          | 7.105e-15      |
| 372          | -2.000e+00     |
| 373          | 1.000e+00      |
| 374          | -4.000e+00     |
| 375          | 1.333e+00      |

Continua na próxima página

| Coordenada y | Valor y (Dual) |
|--------------|----------------|
| 376          | -5.333e+00     |
| 377          | 1.667e+00      |
| 378          | -4.000e+00     |
| 379          | 2.667e+00      |
| 380          | -2.000e+00     |
| 381          | 9.770e-15      |
| 382          | -1.000e+00     |
| 383          | 1.000e+00      |
| 384          | -3.000e+00     |
| 385          | 1.333e+00      |
| 386          | -4.333e+00     |
| 387          | 1.667e+00      |
| 388          | -3.000e+00     |
| 389          | 2.667e+00      |
| 390          | -1.000e+00     |
| 391          | 1.000e+00      |
| 392          | -2.000e+00     |
| 393          | 1.333e+00      |
| 394          | -3.333e+00     |
| 395          | 1.667e+00      |
| 396          | -2.000e+00     |
| 397          | 2.667e+00      |